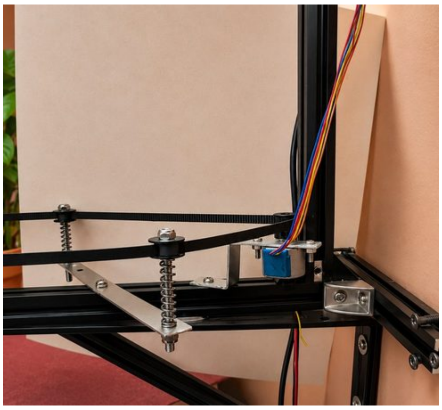
Plant Imaging Rig

IN PROGRESS

Camera housing and diffused light column at the top, motorised platter below carrying the pot.



Spring-loaded idler pulleys press against the drive belt, keeping it under constant tension.

The platter: a lightweight stand with timing belt bonded around the rim to give the drive belt grip, riding on…

PROBLEM

Designed to create an affordable system for capturing long-term plant growth from multiple viewing angles, enabling reconstruction of animated 3D models of slow biological processes. Unlike conventional photogrammetry systems that rely on many synchronized cameras, this approach uses a single rotating camera to reduce cost while supporting experiments lasting days or weeks.

ENGINEERING DECISIONS

- A rotating platform was designed to position the plant accurately for repeatable multi-angle imaging.
- The platter indexes 15° every five minutes and captures a frame at each stop, so a full 360° revolution — 24 angles of the plant — completes every two hours.
- The turntable was machined on a lathe to ensure concentricity, perpendicularity, and smooth rotation, minimizing positional errors during image capture.
- A bearing-supported platform reduces friction and improves rotational stability.
- A spring-loaded belt tensioning mechanism maintains consistent belt tension while minimizing load on the low-torque motor, allowing reliable unattended operation over long periods.
- A custom light diffuser mounted near the camera provides soft, repeatable illumination while reducing harsh reflections.

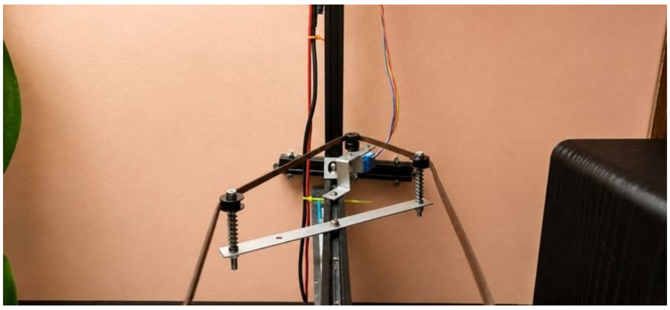
CHALLENGES

- Finding the correct balance between belt tension and motor torque required multiple iterations.
- The rotating platform needed to remain both precisely concentric and perpendicular to the rotation axis to prevent changes in belt tension and unwanted vibration.
- Long-duration experiments demanded reliable mechanical performance with minimal maintenance.

HARDWARE

Raspberry Pi · Camera · Motor Control · Lighting · 3D Printing

Lights off: the camera housing, the extrusion arm carrying the platter, and the belt running to the motor pulley.

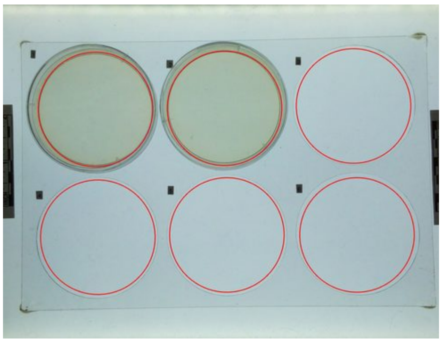


The drive seen from the front: the motor pulley at the rail, a spring-loaded idler on each side, and the belt running down in a V to the platter below.

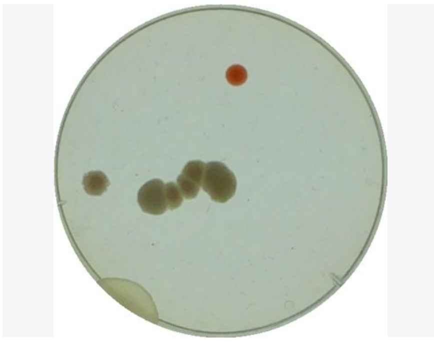
SporeScope — Imaging Rig

2025

2020 extrusion frame with a backlit stage below and the overhead camera under the top plate.



Raw overhead frame from the Raspberry Pi camera



One plate cropped out of that frame and stored as its own record in the database

PROBLEM

Designed to automate long-term monitoring of fungal growth and evaluate whether simple, low-cost backlit imaging is sufficient to distinguish healthy mycelium from contamination without specialized optics.

ENGINEERING DECISIONS

- Built from modular aluminum extrusion to maintain stable camera-to-sample geometry during multi-week experiments.
- Camera distance was intentionally increased beyond the minimum focus distance to reduce lens distortion while maintaining adequate resolution.
- Uniform edge-lit LED panel provides repeatable shadow-based imaging across the entire plate.
- A laser-cut plate holder seats each dish in a numbered position, preventing slipping and making remounting repeatable between runs.
- Settled on a 30-minute capture interval — frequent enough to follow growth, without wasting cloud storage in the AWS S3 database.

CHALLENGES

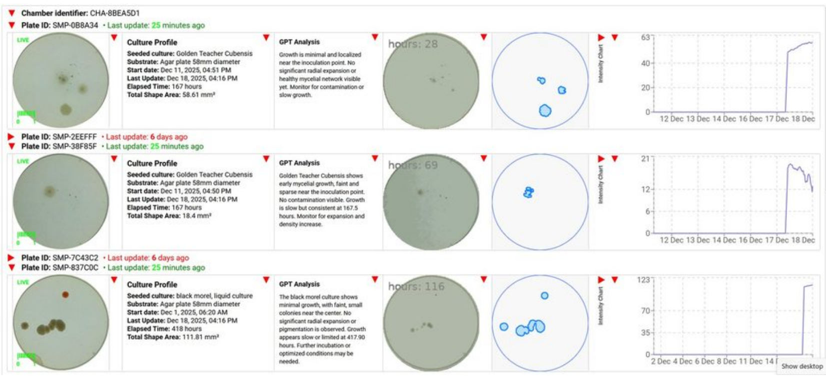
- High humidity caused condensation on agar plates, introducing image noise and reducing measurement reliability.
- Maintaining sub-pixel positional repeatability was essential because image analysis compares the same pixels over time.

OUTCOME

- Raspberry Pi 4 + Camera Module 3 capture time-series images automatically.
- Images are analyzed using both traditional image-processing techniques and an LLM, whose contamination assessments are compared with quantitative measurements.

HARDWARE

Aluminum Extrusion · Raspberry Pi · Backlit Imaging · Biological Imaging · Shape Detection



Web app: one live row per plate

Laser-cut plate holder on the backlit stage

Bio Chart — Sensor & Imaging Chamber

2023

Chamber open with the interior LED lighting on; the Raspberry Pi sits outside on the wall, camera ribbon…

Raw full-size frame from the Pi camera

What the camera produces: one flask cropped out of a capture and stored as its own record

PROBLEM

Developed as a self-contained chamber for long-term observation of mycelium growing inside a closed environment, reducing the need for manual inspection.

ENGINEERING DECISIONS

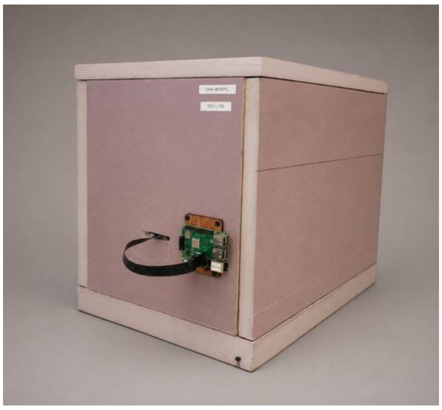
- Fully enclosed chamber minimizes environmental disturbance.
- Integrated lighting ensures repeatable imaging conditions across long experiments.

OUTCOME

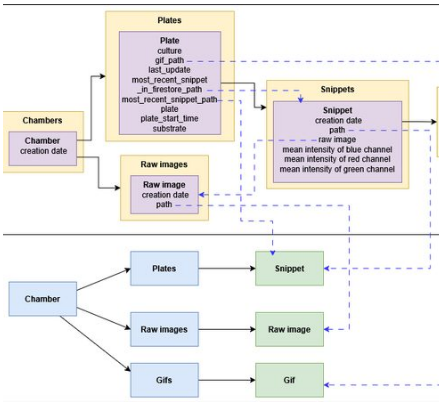
- Raspberry Pi-based imaging station captures photographs every 30 minutes.
- Images are uploaded automatically and processed into an interactive web interface with time-lapse GIF generation.
- RGB color metrics are tracked over time to identify growth trends and detect contamination.
- The system is designed to notify the user via email or SMS when predefined conditions indicate abnormal growth.

HARDWARE

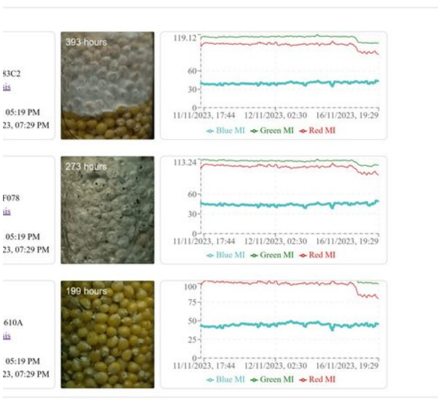
Raspberry Pi · Microcontrollers · Sensors · Image Capture · Data Acquisition · Biological Monitoring



Closed, with the chamber ID and static IP on the front plate.



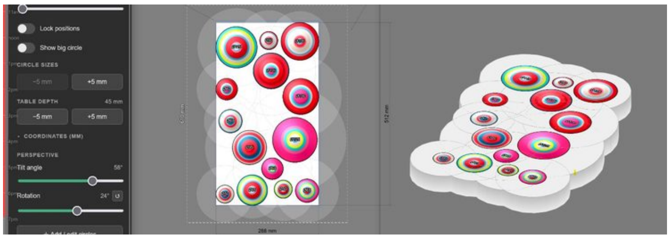
Data model: Firestore holds chambers → plates → snippets → shapes; Firebase Storage mirrors the sam…



Web interface: one row per flask

Shared Signal

IN PROGRESS



The layout tool: circles at real dimensions with live animation, tilt and rotation preview, and CSV export of coordinates.

The tabletop cut to that outline, every circle bored at its planned diameter, wiring routed underneath.

PROBLEM

Developed for an interactive gallery installation where animated light patterns are displayed through hundreds of precisely positioned openings in a tabletop. The project required both a custom physical display system and software tools for rapid design iteration.

ENGINEERING DECISIONS

- Chose a Raspberry Pi 5 to drive a high-resolution LCD display with smooth real-time animation while keeping the hardware compact enough to integrate directly behind the tabletop.
- Mounted the Raspberry Pi directly to the display to minimize footprint, simplify cable management, and keep the installation self-contained while maintaining physical access for maintenance.
- The installation runs locally during exhibition without requiring an Internet connection. Content is developed on a web-based interface, deployed to the Raspberry Pi, and executed entirely from local storage.

CUSTOM DESIGN TOOLING

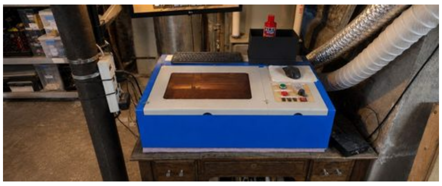
- Developed a custom interactive design application after determining that traditional CAD software was too slow for iterative visual exploration.
- The application automatically generates and updates hundreds of circular cutouts while providing immediate rendered previews from multiple viewing angles, allowing rapid experimentation with layout and animation before fabrication.

HARDWARE

Fabrication · Raspberry Pi · LCD · Geometry

Laser Cutting & Engraving

ONGOING



The machine in my home workshop

Engraved name plate marking what a storage shelf holds.

Plates 2.3 and 3.2

PROBLEM

A collection of small practical tools, labels, fixtures, and organizers created to solve everyday workflow problems around the house and workshop.

ENGINEERING DECISIONS

Rather than temporary labels or tape, laser engraving creates durable markings that remain readable throughout the lifetime of the object.

HARDWARE

Laser Cutting · Engraving · Plywood · Shop Fixtures

3D-Printed Coffee Dosing Cone

FINISHED · IN USE

The cone printed in flexible TPU, stained from daily use.

Seated on the 58 mm portafilter, gripping the basket rim by friction.

PROBLEM

Designed after finding commercial metal dosing funnels inconvenient for daily use.

ITERATION

- Multiple versions explored height, taper angle, flexibility, and material selection.
- Final design uses TPU for a secure friction fit and easier one-handed removal.

OUTCOME

- Used daily, providing continuous real-world validation.

HARDWARE

CAD · 3D Printing · TPU · Functional Design